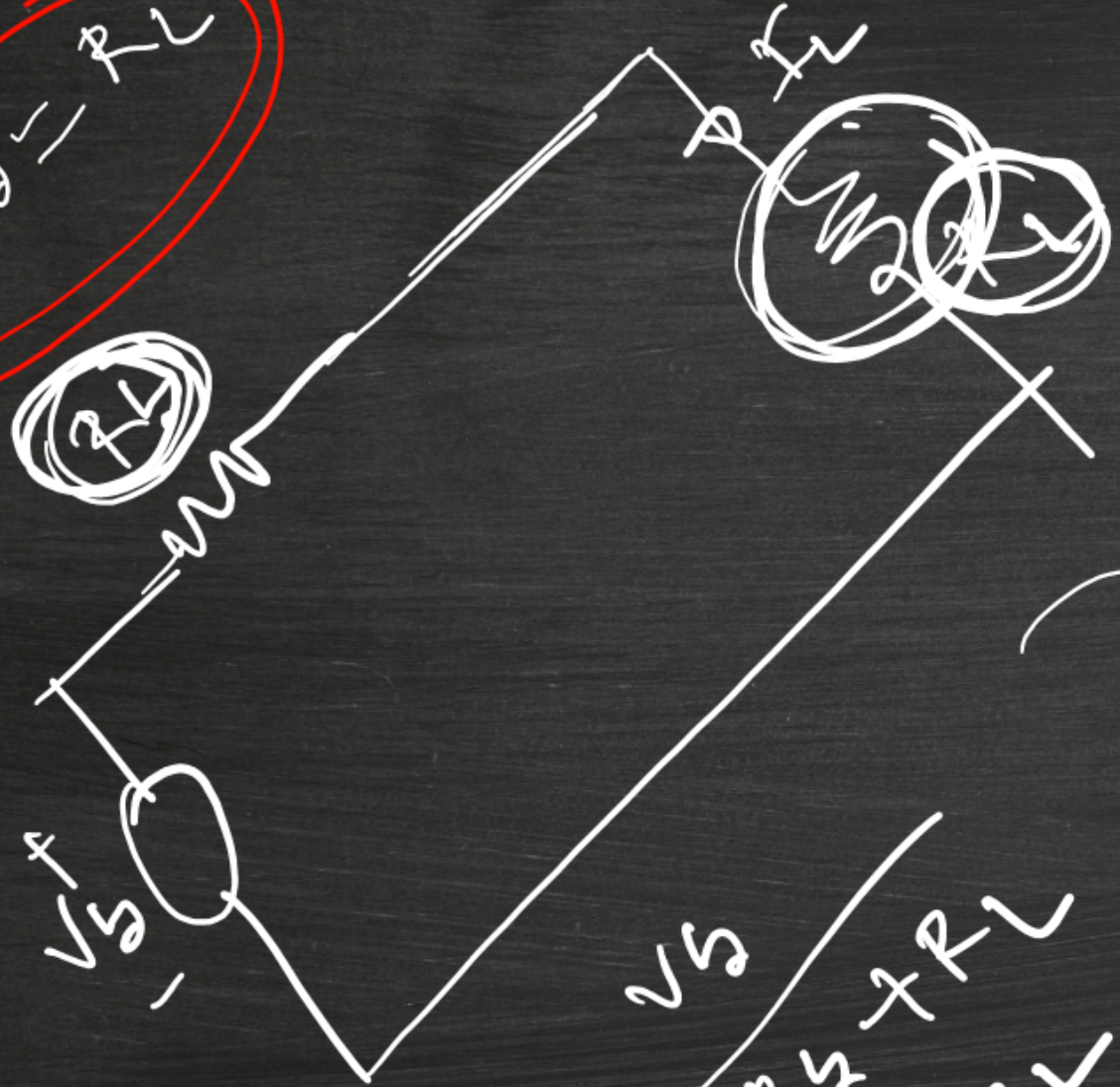




Superposition

What's AP?

$R_S = R_L$



For max



$I_L =$

$\frac{V_S}{R_S + R_L}$

$\times R_L$

$= \frac{V_S \times R_L}{R_S + R_L}$

$\Rightarrow \frac{d}{dR_L} \left(\frac{V_S \times R_L}{R_S + R_L} \right) = 0$

$\Rightarrow \frac{d}{dR_L} (P_L) = 0$

$\Rightarrow \frac{V_S^2 \times R_L}{(R_S + R_L)^2} = 0$

$\Rightarrow \frac{V_S^2 \times (R_S + R_L) - 2R_L \times V_S^2}{(R_S + R_L)^3} = 0$

$\Rightarrow \frac{V_S^2 \times (R_S + R_L - 2R_L)}{(R_S + R_L)^3} = 0$

$\Rightarrow \frac{V_S^2 \times (R_S - R_L)}{(R_S + R_L)^3} = 0$

$\Rightarrow R_S - R_L = 0$

$\Rightarrow R_S + R_L = 2R_L$

$\Rightarrow R_S = 2R_L - R_L$

$\Rightarrow R_S = R_L$

$$\frac{d}{dx}(x^9) = 9x^8$$

$$\left[\frac{d}{dx} \left(\frac{1}{x} \right) \right] =$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx} (x^2 + 12x) = 2x + 12$$

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx} (x^2 + 12x) = 2x + 12$$

$$\frac{d}{dx} (x^2) = 2x$$

$$\frac{d}{dx} (x) = 1$$

$$\frac{d}{dx} =$$

$$\frac{d}{dx} (x^2) = 2x$$

$$\frac{d}{dx} (x^2 + x) = 2x + 1$$

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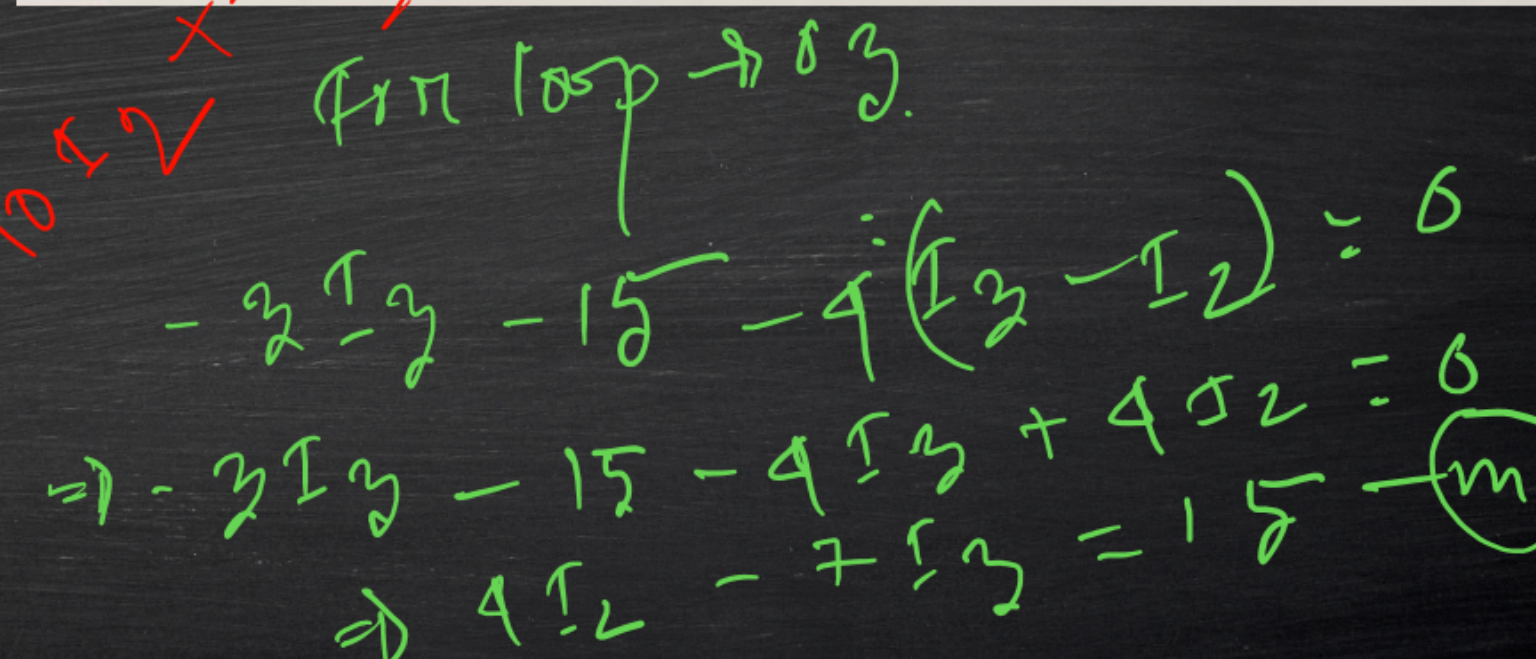
$$\frac{d}{dx} (x^2 + x) = 2x + 1$$

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SOFTMAX
 ONLINE SCHOOL

For loop $\rightarrow$ 01
 12 - 4I1 - 3I2 = 0
 4I1 - 3I2 = 0
 For loop $\rightarrow$ 02
 5I2 - 4I1 - 3I2 = 0
 2I2 - 4I1 = 0
 I2 = 2I1
 For loop $\rightarrow$ 03
 6I3 - 5I2 - 4I1 = 0
 6I3 - 5(2I1) - 4I1 = 0
 6I3 - 10I1 - 4I1 = 0
 6I3 - 14I1 = 0
 I3 = 14/6 I1 = 7/3 I1
 For loop $\rightarrow$ 04
 7I4 - 6I3 - 5I2 = 0
 7I4 - 6(7/3 I1) - 5(2I1) = 0
 7I4 - 14I1 - 10I1 = 0
 7I4 - 24I1 = 0
 I4 = 24/7 I1 = 24/7 I1
 For loop $\rightarrow$ 05
 8I5 - 7I4 - 6I3 = 0
 8I5 - 7(24/7 I1) - 6(7/3 I1) = 0
 8I5 - 24I1 - 14I1 = 0
 8I5 - 38I1 = 0
 I5 = 38/8 I1 = 19/4 I1
 For loop $\rightarrow$ 06
 9I6 - 8I5 - 7I4 = 0
 9I6 - 8(19/4 I1) - 7(24/7 I1) = 0
 9I6 - 38I1 - 24I1 = 0
 9I6 - 62I1 = 0
 I6 = 62/9 I1 = 62/9 I1
 For loop $\rightarrow$ 07
 10I7 - 9I6 - 8I5 = 0
 10I7 - 9(62/9 I1) - 8(19/4 I1) = 0
 10I7 - 62I1 - 38I1 = 0
 10I7 - 100I1 = 0
 I7 = 10I1
 For loop $\rightarrow$ 08
 11I8 - 10I7 - 9I6 = 0
 11I8 - 10(10I1) - 9(62/9 I1) = 0
 11I8 - 100I1 - 62I1 = 0
 11I8 - 162I1 = 0
 I8 = 162/11 I1 = 162/11 I1
 For loop $\rightarrow$ 09
 12I9 - 11I8 - 10I7 = 0
 12I9 - 11(162/11 I1) - 10(10I1) = 0
 12I9 - 162I1 - 100I1 = 0
 12I9 - 262I1 = 0
 I9 = 262/12 I1 = 131/6 I1
 For loop $\rightarrow$ 10
 13I10 - 12I9 - 11I8 = 0
 13I10 - 12(131/6 I1) - 11(162/11 I1) = 0
 13I10 - 262I1 - 162I1 = 0
 13I10 - 424I1 = 0
 I10 = 424/13 I1 = 424/13 I1
 For loop $\rightarrow$ 11
 14I11 - 13I10 - 12I9 = 0
 14I11 - 13(424/13 I1) - 12(131/6 I1) = 0
 14I11 - 424I1 - 262I1 = 0
 14I11 - 686I1 = 0
 I11 = 686/14 I1 = 49I1
 For loop $\rightarrow$ 12
 15I12 - 14I11 - 13I10 = 0
 15I12 - 14(49I1) - 13(424/13 I1) = 0
 15I12 - 686I1 - 424I1 = 0
 15I12 - 1110I1 = 0
 I12 = 1110/15 I1 = 74I1
 For loop $\rightarrow$ 13
 16I13 - 15I12 - 14I11 = 0
 16I13 - 15(74I1) - 14(49I1) = 0
 16I13 - 1110I1 - 686I1 = 0
 16I13 - 1796I1 = 0
 I13 = 1796/16 I1 = 449/4 I1
 For loop $\rightarrow$ 14
 17I14 - 16I13 - 15I12 = 0
 17I14 - 16(449/4 I1) - 15(74I1) = 0
 17I14 - 1796I1 - 1110I1 = 0
 17I14 - 2906I1 = 0
 I14 = 2906/17 I1 = 2906/17 I1
 For loop $\rightarrow$ 15
 18I15 - 17I14 - 16I13 = 0
 18I15 - 17(2906/17 I1) - 16(449/4 I1) = 0
 18I15 - 2906I1 - 1796I1 = 0
 18I15 - 4702I1 = 0
 I15 = 4702/18 I1 = 2351/9 I1
 For loop $\rightarrow$ 16
 19I16 - 18I15 - 17I14 = 0
 19I16 - 18(2351/9 I1) - 17(2906/17 I1) = 0
 19I16 - 4702I1 - 2906I1 = 0
 19I16 - 7608I1 = 0
 I16 = 7608/19 I1 = 7608/19 I1
 For loop $\rightarrow$ 17
 20I17 - 19I16 - 18I15 = 0
 20I17 - 19(7608/19 I1) - 18(2351/9 I1) = 0
 20I17 - 7608I1 - 4702I1 = 0
 20I17 - 12310I1 = 0
 I17 = 12310/20 I1 = 1231/2 I1
 For loop $\rightarrow$ 18
 21I18 - 20I17 - 19I16 = 0
 21I18 - 20(1231/2 I1) - 19(7608/19 I1) = 0
 21I18 - 12310I1 - 7608I1 = 0
 21I18 - 19918I1 = 0
 I18 = 19918/21 I1 = 19918/21 I1
 For loop $\rightarrow$ 19
 22I19 - 21I18 - 20I17 = 0
 22I19 - 21(19918/21 I1) - 20(1231/2 I1) = 0
 22I19 - 19918I1 - 12310I1 = 0
 22I19 - 32228I1 = 0
 I19 = 32228/22 I1 = 16114/11 I1
 For loop $\rightarrow$ 20
 23I20 - 22I19 - 21I18 = 0
 23I20 - 22(16114/11 I1) - 21(19918/21 I1) = 0
 23I20 - 32228I1 - 19918I1 = 0
 23I20 - 52146I1 = 0
 I20 = 52146/23 I1 = 52146/23 I1
 For loop $\rightarrow$ 21
 24I21 - 23I20 - 22I19 = 0
 24I21 - 23(52146/23 I1) - 22(16114/11 I1) = 0
 24I21 - 52146I1 - 32228I1 = 0
 24I21 - 84374I1 = 0
 I21 = 84374/24 I1 = 42187/12 I1
 For loop $\rightarrow$ 22
 25I22 - 24I21 - 23I20 = 0
 25I22 - 24(42187/12 I1) - 23(52146/23 I1) = 0
 25I22 - 84374I1 - 52146I1 = 0
 25I22 - 136520I1 = 0
 I22 = 136520/25 I1 = 54608/10 I1 = 54608/5 I1
 For loop $\rightarrow$ 23
 26I23 - 25I22 - 24I21 = 0
 26I23 - 25(54608/5 I1) - 24(42187/12 I1) = 0
 26I23 - 546080I1 - 84374I1 = 0
 26I23 - 630454I1 = 0
 I23 = 630454/26 I1 = 315227/13 I1
 For loop $\rightarrow$ 24
 27I24 - 26I23 - 25I22 = 0
 27I24 - 26(315227/13 I1) - 25(54608/5 I1) = 0
 27I24 - 630454I1 - 546080I1 = 0
 27I24 - 1176534I1 = 0
 I24 = 1176534/27 I1 = 435753/10 I1 = 435753/10 I1
 For loop $\rightarrow$ 25
 28I25 - 27I24 - 26I23 = 0
 28I25 - 27(435753/10 I1) - 26(315227/13 I1) = 0
 28I25 - 1176534I1 - 630454I1 = 0
 28I25 - 1806988I1 = 0
 I25 = 1806988/28 I1 = 451747/7 I1
 For loop $\rightarrow$ 26
 29I26 - 28I25 - 27I24 = 0
 29I26 - 28(451747/7 I1) - 27(435753/10 I1) = 0
 29I26 - 1806988I1 - 1176534I1 = 0
 29I26 - 2983522I1 = 0
 I26 = 2983522/29 I1 = 2983522/29 I1
 For loop $\rightarrow$ 27
 30I27 - 29I26 - 28I25 = 0
 30I27 - 29(2983522/29 I1) - 28(451747/7 I1) = 0
 30I27 - 2

Node Method

App Super node

$$V_1 + 3 - V_2 = 0$$

$$\Rightarrow V_1 - V_2 = -3$$

App Node 1

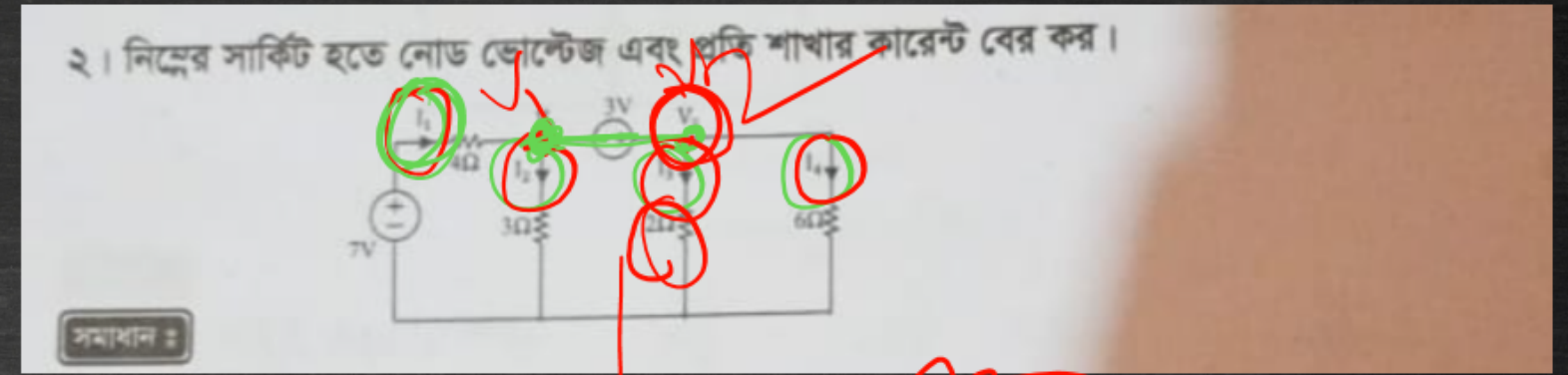
$$I_1 = I_2 + I_3 + I_4$$

$$\Rightarrow 0 - \frac{V_1}{3} = \frac{V_1}{3} + \frac{V_1}{2} + \frac{V_2}{6}$$

$$\Rightarrow -\frac{V_1}{3} = \frac{2V_1 + 3V_1 + V_2}{6}$$

$$\Rightarrow -2V_1 = 5V_1 + V_2$$

$$\Rightarrow -7V_1 - V_2 = 0$$



Super Node

Direction given

KCL

$\downarrow$

$\frac{V_1}{3} + \frac{V_2}{6} - V_2 = 0$

